Vehicle Steering Control: PID and Lead-Lag Design for Autonomous Lane Keeping

Dhruv Choudhary
Department of Electrical and Computer Engineering
University of California San Diego, La Jolla, CA 92093
ECE 171A — Linear Control System Theory, Spring 2026

***Abstract*—This paper presents the design and analysis of a lane-keeping controller for the single-track bicycle model of a passenger car traveling at $v = 25$ m/s. A Routh-Hurwitz analysis shows that pure proportional feedback of the lane error is conditionally stable and essentially undamped, motivating phase-lead compensation. Two controllers are designed and compared: a PID compensator tuned by loop shaping at $\omega_g = 12$ rad/s (phase margin 66.0°, bandwidth 17.6 rad/s), and a root-locus-designed lead-lag compensator (phase margin 60.9°, bandwidth 18.7 rad/s). Both achieve rise times under 100 ms and track a $0.32\,g$ curvature step with less than 6 cm peak lane deviation. The PID achieves zero steady-state lane error through integral action; the lead-lag leaves a 3.7 mm residual offset. Robustness to a 44% reduction in road adhesion is demonstrated for both designs.**

I. Introduction

Automatic lane keeping requires a controller that steers the front wheels to keep the vehicle centered in its lane, given measurements of the lateral deviation from the road reference. This project designs and analyzes such a controller for the single-track ("bicycle") model of a passenger car at constant speed $v = 25$ m/s. The model, originally from Ackermann et al. [3], outputs the lateral acceleration at a virtual sensor point located a distance d_S ahead of the center of gravity. Integrating the difference between that acceleration and the road curvature demand twice gives the lane error ΔY_S, which the controller drives to zero by commanding the front-wheel steering angle.

The project is split into three tasks. First, the open-loop vehicle dynamics are characterized through the state-space model, transfer function, pole-zero map, step response, and Bode plot. Second, a PID controller is designed: a proportional-only loop is shown (by Routh-Hurwitz analysis and simulation) to be conditionally stable and essentially undamped, so derivative action is added for damping and integral action eliminates the steady-state lane offset on curves. Third, a lead-lag compensator is designed using the root locus, where a lead zero pulls the poorly damped branches near the imaginary axis deep into the left half-plane, and a lag stage recovers low-frequency gain. Both designs are verified against the frequency response and compared in simulation.

II. Vehicle Model and Control Architecture

A. State-Space Model and Transfer Function

The lateral dynamics are described by the side-slip angle β and the yaw rate $\dot{\psi}$, with the front-wheel steering angle u as input:

TABLE I: Model parameters

v (m/s)	M (kg)	J (kg m^2)	ℓ_f (m)	ℓ_r (m)	c_f=c_r (N/rad)
25	1573	2873	1.10	1.58	80000
μ	d_S (m)				
0.9	1.96				

$$\dot{x} = Ax + Bu, \quad \ddot{y}_S = Cx + Du, \quad x = \begin{bmatrix} \beta \\ \dot{\psi} \end{bmatrix}. \tag{1}$$

Substituting the model parameters from Table I gives

$$A = \begin{bmatrix} -3.6618 & -0.9648 \\ 12.0292 & -3.7154 \end{bmatrix}, \; B = \begin{bmatrix} 1.8309 \\ 27.5670 \end{bmatrix}, \\ C = \begin{bmatrix} -67.9675 & 0.8788 \end{bmatrix}, \; D = 99.8037. \tag{2}$$

The transfer function from steering angle to sensor-point lateral acceleration, $G(s) = C(sI - A)^{-1}B + D$, is

$$G(s) = \frac{99.804\,s^2 + 636.06\,s + 3969.71}{s^2 + 7.3772\,s + 25.2115}. \tag{3}$$

Two features of (3) matter for control design. The system is *biproper* (relative degree zero): the direct feedthrough $D = 99.8$ produces an instantaneous lateral acceleration response to steering, and the Bode magnitude approaches $20\log_{10}(99.8) = 40.0$ dB at high frequency. The poles are at $s = -3.689 \pm 3.407j$ ($\omega_n = 5.02$ rad/s, $\zeta = 0.735$) and the zeros at $s = -3.187 \pm 5.443j$ ($\omega_n = 6.31$ rad/s, $\zeta = 0.505$).

B. Closed-Loop Architecture

The control architecture is shown in Fig. 1. The road demands a reference lateral acceleration $\ddot{y}_{\mathrm{ref}} = v^2 \rho_{\mathrm{ref}}$. Integrating the difference between achieved and demanded accelerations twice produces the lane error ΔY_S, fed back to the controller $F(s)$ with a sign flip:

$$\Delta Y_S(s) = \frac{\ddot{y}_S(s) - \ddot{y}_{\mathrm{ref}}(s)}{s^2}, \quad u(s) = F(s)\big(-\Delta Y_S(s)\big). \tag{4}$$

Closing the loop gives

$$\ddot{y}_S = T(s)\,\ddot{y}_{\mathrm{ref}}, \quad T(s) = \frac{L(s)}{1 + L(s)}, \quad L(s) = \frac{F(s)G(s)}{s^2}, \tag{5}$$

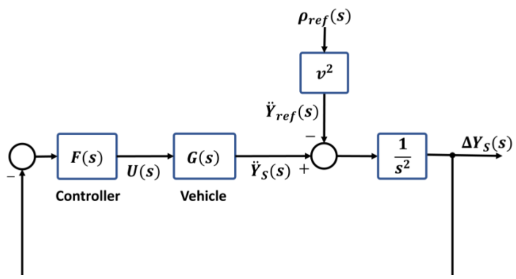


Fig. 1: Closed-loop architecture. The plant seen by the controller is $P(s) = G(s)/s^2$.

so the plant seen by the controller is $P(s) = G(s)/s^2$, and the steady-state lane error for a constant curvature step a_0 satisfies

$$\Delta Y_{S,ss} = \lim_{s\to 0} s \cdot \frac{-a_0/s}{s^2 + G(s)F(s)} = \frac{-a_0}{G(0)\,F(0)}. \tag{6}$$

Offset is zero only if $F(0) = \infty$ (integral action) or small if $F(0)$ is large but finite (lag compensation). The benchmark reference is a curve of radius $R = 200$ m:

$$\ddot{y}_{\text{ref}} = v^2 \rho_{\text{ref}} = 25^2 \cdot \tfrac{1}{200} = 3.125 \text{ m/s}^2 \tag{7}$$

($0.32\,g$), which requires a steady steering angle of $a_0/G(0) = 3.125/157.46 = 0.0198$ rad $= 1.14°$.

III. Open-Loop Analysis

Script `p1_open_loop.m` constructs the model with `ss` and `tf` and extracts poles, zeros, and gain with `zpkdata` (the project description lists this as `pzkdata`; the correct toolbox name is `zpkdata`).

The pole-zero map (Fig. 2) shows both pairs are complex and firmly in the left half-plane: the vehicle is inherently stable at this speed. The DC gain $G(0) = 3969.71/25.21 = 157.46$ (m/s^2)/rad means 1° of steering at 25 m/s produces 2.75 m/s^2 of steady lateral acceleration.

The step response (Fig. 3) confirms the biproper structure: the output jumps to $D = 99.8$ m/s^2 at $t = 0^+$, then settles to $G(0) = 157.5$ m/s^2 with 1.3% overshoot, rise time 0.47 s, and 2% settling time 0.66 s.

The Bode plot (Fig. 4) is unusually flat (43.9 dB at low frequency, dipping to $\approx$ 38 dB near 6 rad/s, recovering to 40.0 dB at high frequency) because the pole and zero resonances nearly cancel each other's phase contributions. $G(s)$ supplies no integration and little net phase shaping; all meaningful loop dynamics come from the $1/s^2$ of the architecture and from the controller.

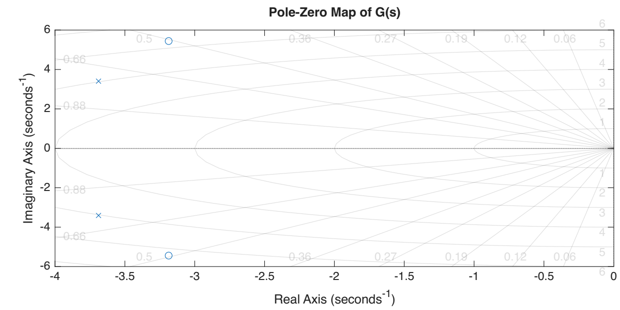


Fig. 2: Pole-zero map of $G(s)$. Both pairs are complex and in the left half-plane.

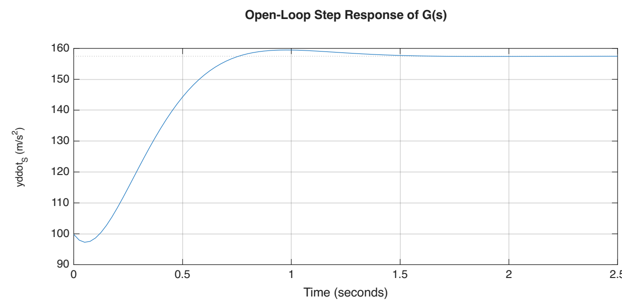


Fig. 3: Open-loop step response of $G(s)$ (1 rad input).

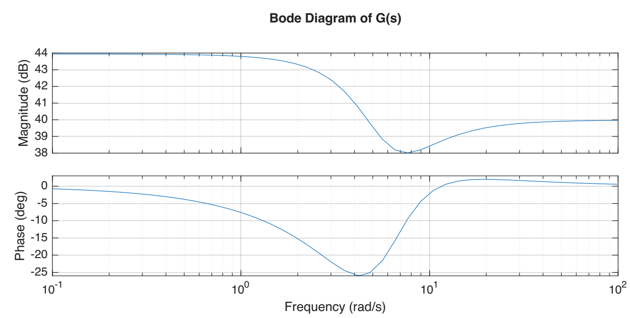


Fig. 4: Bode diagram of $G(s)$. The system is biproper so the magnitude does not roll off at high frequency.

IV. PID Controller Design

A. Proportional Control: Conditional Stability

With $F(s) = k_p$, the closed-loop characteristic polynomial is

$$\Delta(s) = s^4 + 7.3772\,s^3 + (25.2115 + 99.804\,k_p)\,s^2 + 636.06\,k_p\,s + 3969.71\,k_p. \tag{8}$$

The Routh table (Table II) shows the loop is stable if and only if

$$8640.4\,k_p - 13249.4 > 0 \implies k_p > 1.5334. \tag{9}$$

At the critical gain $k_p^* = 1.5334$ the s^2-row auxiliary polynomial gives the sustained-oscillation frequency

$$\omega^2 = \frac{3969.71\,k_p^*}{25.2115 + 13.5843\,k_p^*} = 132.2 \implies \omega = 11.50 \text{ rad/s}. \tag{10}$$

TABLE II: Routh table for proportional-only control

s^4	1	$25.21 + 99.80\,k_p$	$3969.71\,k_p$
s^3	7.3772	$636.06\,k_p$	
s^2	$25.21 + 13.58\,k_p$	$3969.71\,k_p$	
s^1	$\dfrac{k_p(8640.4\,k_p - 13249.4)}{25.21 + 13.58\,k_p}$	0	
s^0	$3969.71\,k_p$	0	

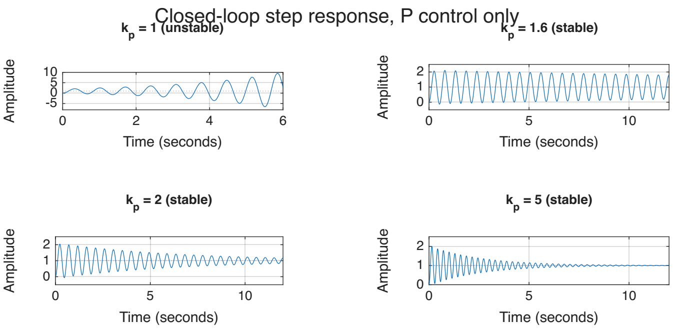


Fig. 5: Closed-loop step responses with proportional control only. $k_p = 1$ is unstable ($k_p < 1.5334$); larger gains are stable but practically undamped.

Both results are confirmed numerically in `p2_pid.m`. The loop is *conditionally stable*: reducing the gain destabilizes it. Worse, even well above the critical gain the dominant poles remain nearly undamped (at $k_p = 5$: $-0.393 \pm 21.85j$, $\zeta = 0.018$), as visible in Fig. 5. The root-locus explanation is in Section V: the two branches from the double integrator can never move far from the imaginary axis under pure gain feedback.

B. Adding Derivative Action

A PD controller $F(s) = k_p + k_d s/(T_f s + 1)$ (filtered at $T_f = 0.01$ s to keep the controller proper) adds up to $+90°$ of phase near crossover. With $k_p = 0.50$ and $k_d = 0.13$ the loop achieves a 66° phase margin with rise time 93 ms, overshoot 23.5%, and settling 1.42 s. However, by (6) the steady-state lane error on any constant-radius curve is $\Delta Y_{S,ss} = -3.125/(157.46 \times 0.50) = -3.97$ cm (Fig. 6). Integral action is needed.

C. Final PID Design

The full controller in the parallel form used by MATLAB's `pid` command is

$$F_{\text{PID}}(s) = k_p + \frac{k_i}{s} + \frac{k_d\, s}{T_f s + 1} \tag{11}$$

with $k_p = 0.50$, $k_i = 0.40$, $k_d = 0.13$, $T_f = 0.01$ s. The gains were chosen by loop shaping at $\omega_g = 12$ rad/s, following the procedure from lecture:

1) **Crossover:** $\omega_g = 12$ rad/s sits above the marginal frequency (10) so the phase has recovered, and keeps the bandwidth high enough for a sub-100 ms rise time while staying a decade below the derivative filter pole at $1/T_f = 100$ rad/s.
2) **Zeros:** Writing the unfiltered PID as $F_{\text{PID}}(s) \approx k_d(s + z_1)(s + z_2)/s$ with $z_1 = 1$, $z_2 = 3$ rad/s, the controller contributes $\arctan(12) + \arctan(4) - 90° \approx +71.2°$ of phase at crossover, and $|G(j12)| = 87.1$ is read from the Bode plot.
3) **Gain:**

$$k_d \frac{|j12 + 1|\,|j12 + 3|}{12} \cdot \frac{|G(j12)|}{12^2} = 1 \\ \Rightarrow\ k_d \approx 0.133. \tag{12}$$

After rounding and a final tuning pass the gains in (11) place the crossover at exactly 12.0 rad/s.

D. Margins, Step Response, and Bandwidth

Fig. 7 shows `margin(L)` for the final design: $\mathbf{PM = 66.0°}$ at 12.0 rad/s, and a gain margin of -21.1 dB at 2.57 rad/s. The negative decibel gain margin is not a bug: because the loop is conditionally stable the $-180°$ phase crossing occurs while $|L|$ is still 21.1 dB *above* unity, so `margin` reports the factor ($0.088\times$) by which the gain may be *reduced* before the loop destabilizes. The design tolerates a 91% loss of loop gain or an arbitrary gain increase (the phase never crosses $-180°$ again at high frequency).

Fig. 8 gives the closed-loop step response of $T(s)$ and the closed-loop magnitude with the -3 dB bandwidth: rise time 94 ms, overshoot 25.8%, settling time 1.44 s, bandwidth 17.6 rad/s.

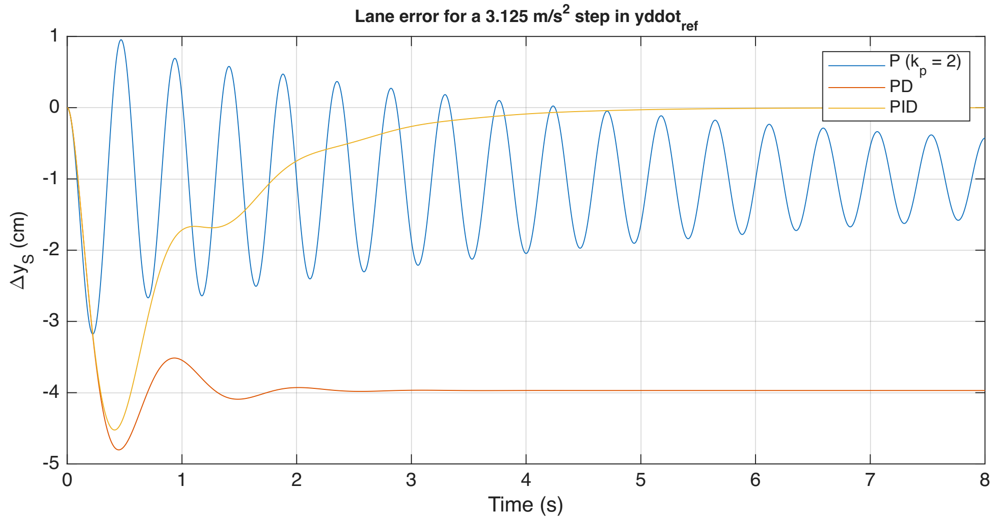


Fig. 6: P vs. PD vs. PID: lane error for the 3.125 m/s^2 reference step. P rings indefinitely, PD leaves a -4 cm steady offset, PID drives the error to zero.

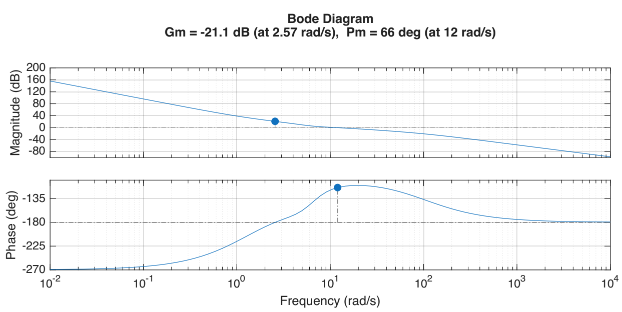
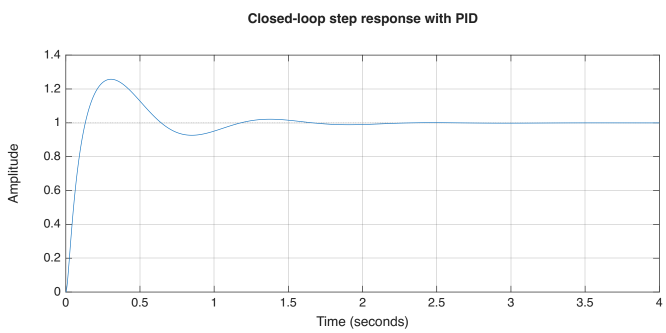
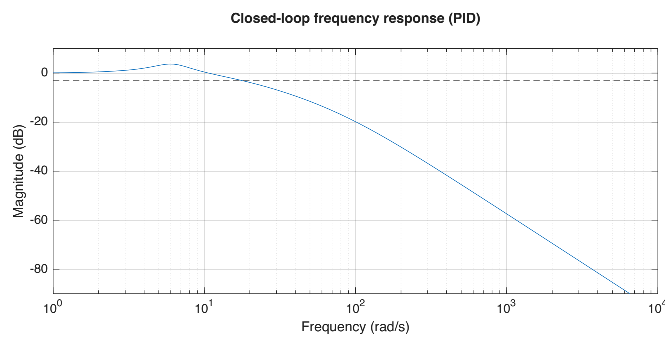


Fig. 7: Loop gain margins with the PID controller. The GM is a gain-reduction margin because the loop is conditionally stable.



(a) Closed-loop step response.



(b) Closed-loop magnitude (-3 dB bandwidth = 17.6 rad/s).

Fig. 8: Final PID design: rise time 94 ms, overshoot 25.8%, settling 1.44 s.

V. Root Locus Analysis and Lead-Lag Design

A. Root Locus of the Uncompensated Loop

Fig. 9 shows `rlocus(G/s^2)`. The loop has $n = 4$ poles (double integrator plus vehicle pair at $-3.69 \pm 3.41j$) and $m = 2$ zeros (vehicle pair at $-3.19 \pm 5.44j$), so $n - m = 2$ branches go to infinity along asymptotes at $\pm 90°$ with centroid

$$\alpha_c = \frac{\sum p_i - \sum z_i}{n - m} = \frac{-7.3772 - (-6.3731)}{2} = -0.502. \tag{13}$$

The two branches from the double integrator bow into the right half-plane and re-cross the imaginary axis at $\pm j11.50$ when $k = 1.533$ (`rlocfind(P, 11.5j)` confirms this, matching (9)–(10) exactly). After re-entering, the branches follow the vertical asymptotes at $\mathrm{Re}(s) = -0.50$, so as $k \to \infty$ the dominant pair keeps a real part near -0.5 while its imaginary part grows without bound: the damping ratio tends to zero, explaining the undamped responses in Fig. 5. No pure-gain controller can fix this.

B. Lead Compensator

A zero at $z = 2$ (just left of the centroid) attracts the branches from the origin deep into the left half-plane. A far pole at $p = 45$ keeps the compensator proper. The same choice gives a maximum phase lead of

$$\begin{aligned} \varphi_{\max} &= \arcsin \frac{1-\alpha}{1+\alpha} = \arcsin \frac{1 - 2/45}{1 + 2/45} = 66.2°, \\ \omega_m &= \sqrt{zp} = \sqrt{90} = 9.49 \text{ rad/s}, \end{aligned} \tag{14}$$

placed just below the intended crossover so most of the $66°$ is available as phase margin.

C. Lag Compensator

The lead stage alone has DC gain $F(0) = K \cdot z/p = 6 \times 2/45 = 0.267$, giving, by (6), a steady-state lane error of -74 mm on the $R = 200$ m curve. A lag stage $(s+1)/(s+0.05)$ multiplies the DC gain by 20 while costing only $\arctan(11.3/1) - \arctan(11.3/0.05) = -4.8°$ of phase near crossover:

$$\begin{aligned} &F(0) = 0.267 \text{ (lead only)} \;\Rightarrow\; \Delta Y_{S,ss} = -74 \text{ mm}; \\ &F(0) = 5.33 \text{ (with lag)} \;\Rightarrow\; \Delta Y_{S,ss} = -3.7 \text{ mm}. \end{aligned} \tag{15}$$

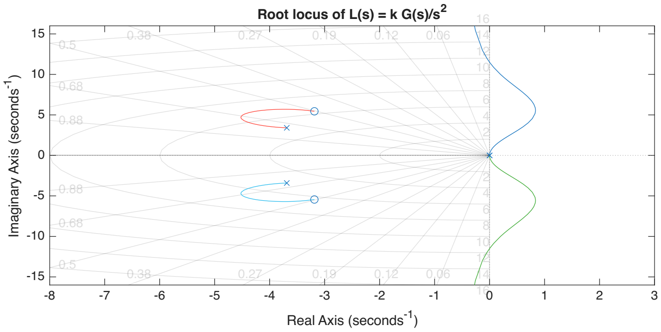
Placing the lag pole at -0.05 (rather than at the origin) makes this a genuinely different design from the PID while leaving a small, quantifiable residual offset to discuss.

D. Gain Selection and Verification

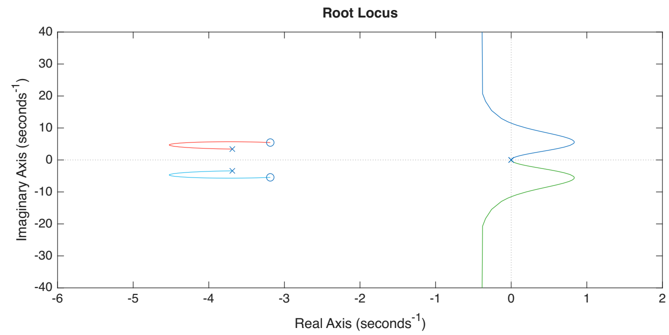
The complete compensator is

$$F_{ll}(s) = 6\,\frac{s+2}{s+45} \cdot \frac{s+1}{s+0.05}. \tag{16}$$

Fig. 10 shows the compensated root locus. The gain was picked where the dominant pair has its best damping, confirmed with `rlocfind(C1*P, -2.07+5.77j)` which returns $K \approx 6$, placing the dominant pair at $-2.07 \pm 5.77j$ ($\zeta = 0.34$, $\omega_n = 6.13$ rad/s). Fig. 11 shows margin verification: $\mathbf{PM = 60.9°}$ at 11.3 rad/s, gain-reduction margin -24.8 dB at 1.88 rad/s. Fig. 12 confirms time-domain performance: rise 93 ms, overshoot 26.1%, settling 1.45 s, bandwidth 18.7 rad/s.

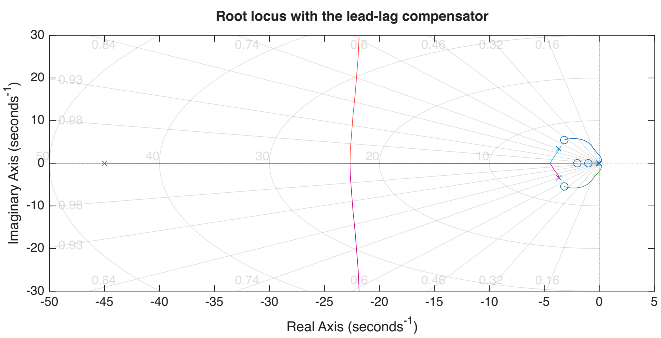


(a) Full view.

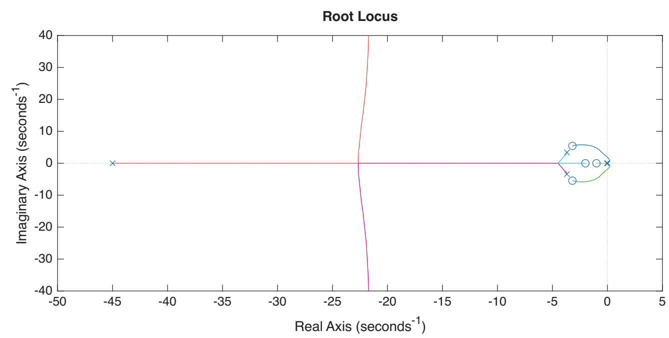


(b) Zoom near the origin: branches bow into the RHP and return at $k = 1.533$, $\omega = 11.5$ rad/s.

Fig. 9: Root locus of $L(s) = kG(s)/s^2$.



(a) Full view.



(b) Zoom: dominant branches pulled into the LHP at $K = 6$.

Fig. 10: Root locus with the lead-lag compensator.

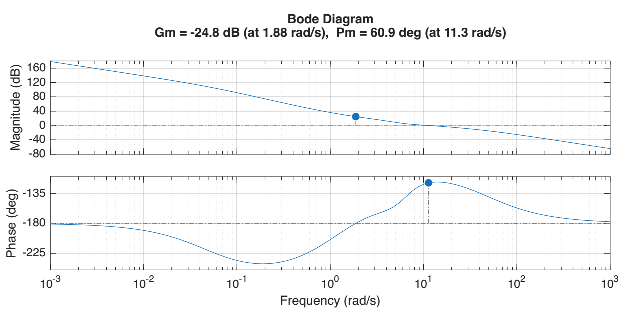
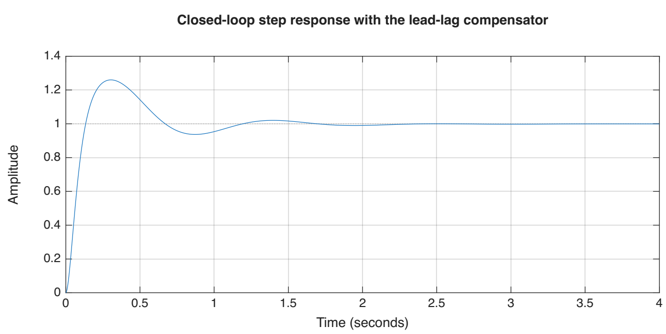
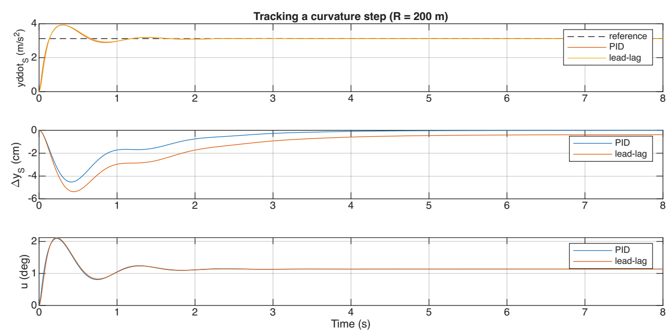


Fig. 11: Loop gain margins with the lead-lag compensator.



(a) Closed-loop step response.



(b) Lane error for the curvature step showing -3.7 mm residual offset.

Fig. 12: Lead-lag closed-loop performance: rise 93 ms, overshoot 26.1%, settling 1.45 s, bandwidth 18.7 rad/s.

TABLE III: Closed-loop performance of the two final designs

Metric	PID	Lead-lag
Gain crossover ω_g (rad/s)	12.0	11.3
Phase margin	$66.0°$	$60.9°$
Gain margin (reduction)	-21.1 dB @ 2.57 rad/s	-24.8 dB @ 1.88 rad/s
Rise time (10–90%)	94 ms	93 ms
Overshoot	25.8%	26.1%
Settling time (2%)	1.44 s	1.45 s
Closed-loop bandwidth	17.6 rad/s	18.7 rad/s
Peak lane error ($a_0 = 3.125$ m/s^2)	4.5 cm	5.4 cm
Steady-state lane error	0	-3.7 mm
Peak / steady steering angle	$2.09°$ / $1.14°$	$2.12°$ / $1.14°$
Phase margin at $\mu = 0.5$	$47.3°$	$45.5°$

VI. Results and Discussion

A. Performance Comparison

Table III collects the key metrics for both final designs. The two controllers are nearly interchangeable in speed and damping, since both were shaped to cross over near 11–12 rad/s with comparable phase. The PID's structural advantage is the true integrator (zero offset, $5°$ more margin); the lead-lag's advantage is slightly more downward gain margin and bandwidth, and no pole at the origin, which avoids integrator-windup concerns if the steering actuator saturates.

B. Constant Curvature Step

Fig. 12(b) shows both loops driving onto the $R = 200$ m curve. The acceleration reaches 3.125 m/s^2 within a quarter second; steering peaks at $2.1°$ before settling to $1.14°$. The lane error transient stays below 6 cm for both designs.

C. S-Maneuver

For a more demanding test, the chosen maneuver is a 26 s, $\approx$600 m S-curve: curvature ramps to $+1/200$ m^{-1} over 1 s, holds for 7 s, reverses to $-1/200$ m^{-1} over 2 s, holds, and ramps back to zero. Both the reference path and the vehicle path are reconstructed from the curvature via

$$\theta(t) = \int_0^t v\rho \, d\sigma, \quad p_x = \int_0^t v\cos\theta \, d\sigma,$$
$$p_y = \int_0^t v\sin\theta \, d\sigma, \quad \rho \approx \ddot{y}_S / v^2, \tag{17}$$

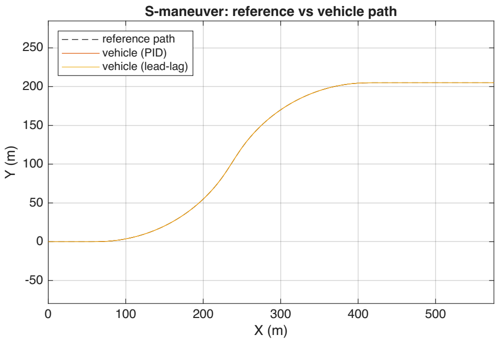
with integrals computed via `cumtrapz`. Fig. 13 shows the reference vs. vehicle paths and the lane error and steering commands over the maneuver. At the scale of the 600 m path the trajectories are indistinguishable. Lane error stays within ± 4.3 cm (PID) and ± 6.0 cm (lead-lag), with peaks at the curvature reversals where $\ddot{y}_{\mathrm{ref}}$ swings by 6.25 m/s^2; steering stays within $\pm 1.35°$.

D. Robustness to Reduced Road Adhesion

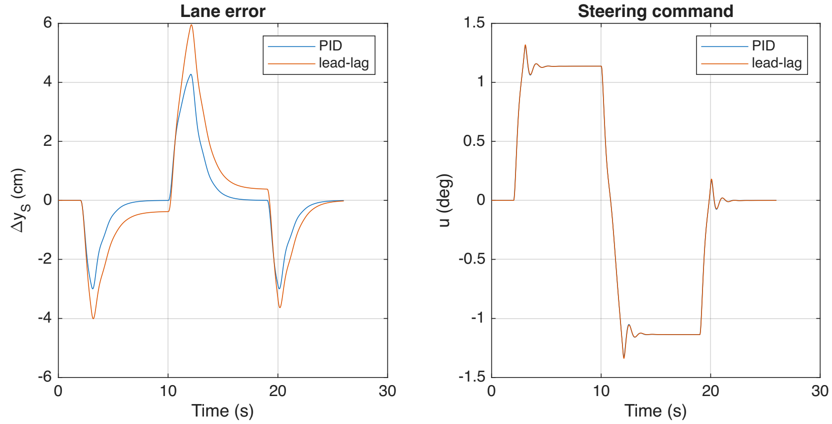
Rebuilding the plant with $\mu = 0.5$ (wet road) and re-evaluating both loops without retuning: both remain stable with phase margins of $47.3°$ (PID) and $45.5°$ (lead-lag) at crossovers near 6 rad/s. Fig. 14 shows the step responses are slower and more oscillatory (overshoot $\approx 42\%$, settling ≈ 3.7 s) but well-behaved. The large gain-reduction margins absorb this 44% loss of plant gain.

E. Limitations

Three caveats bound these results. First, both loops are conditionally stable; the $\approx$ 21–25 dB reduction margins are generous but any component that strongly attenuates the loop at low frequency could destabilize rather than merely slow the system. Second, the $\approx 26\%$ overshoot in acceleration tracking is structural: the fixed error-feedback architecture cannot cancel the closed-loop zeros contributed by the controller; a two-degree-of-freedom structure with feedforward $u_{ff} = \ddot{y}_{\mathrm{ref}}/G(0)$ would remove most of it. Third, the linear single-track model assumes small slip angles and constant speed; gain scheduling on v and μ [3] would be the natural extension.



(a) Reference vs. vehicle paths (overlapping at this scale).



(b) Lane error and steering command over the 26 s maneuver.

Fig. 13: S-maneuver: 26 s, ≈ 600 m double lane change.

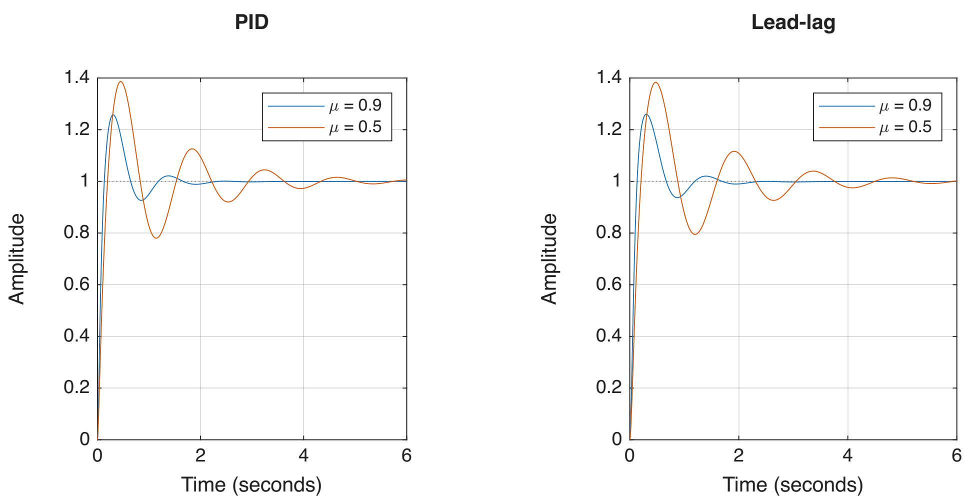


Fig. 14: Closed-loop step responses on dry ($\mu = 0.9$) and wet ($\mu = 0.5$) road, without retuning.

VII. Conclusion

Starting from the single-track vehicle model, this project showed by Routh-Hurwitz and root-locus arguments that pure proportional control of the lane error is conditionally stable and essentially undamped. A loop-shaped PID ($k_p = 0.50$, $k_i = 0.40$, $k_d = 0.13$, $T_f = 0.01$ s) delivers a 66.0° phase margin, 94 ms rise time, 17.6 rad/s bandwidth, and zero steady-state lane error. A root-locus-designed lead-lag, $6(s+2)/(s+45) \cdot (s+1)/(s+0.05)$, delivers comparable speed with a 60.9° margin and a 3.7 mm residual offset. Both track the $0.32\,g$ curvature step with under 6 cm of peak lane deviation and barely 2° of steering, follow a 600 m S-maneuver with centimeter accuracy, and survive a wet-road plant change with $\approx$46° of margin to spare. Within the linear model and the fixed error-feedback architecture either design meets the goals of the project; the PID is the preferred design for its exact curve-holding and slightly larger phase margin.

References

[1] K. J. Åström and R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*, 2nd ed. Princeton, NJ: Princeton University Press, 2021.

[2] N. Atanasov, B. Sankaran, J. Le Ny, G. Pappas, and K. Daniilidis, “Nonmyopic View Planning for Active Object Classification and Pose Estimation,” *IEEE Trans. on Robotics*, vol. 30, no. 5, pp. 1078–1090, 2014.

[3] J. Ackermann, J. Guldner, W. Sienel, R. Steinhauser, and V. I. Utkin, “Linear and nonlinear controller design for robust automatic steering,” *IEEE Trans. on Control Systems Technology*, vol. 3, no. 1, pp. 132–143, 1995.

[4] G. F. Franklin, J. D. Powell, and A. Emami-Naeini, *Feedback Control of Dynamic Systems*, 8th ed. New York, NY: Pearson, 2019.

[5] R. Rajamani, *Vehicle Dynamics and Control*, 2nd ed. New York, NY: Springer, 2012.

[6] The MathWorks, Inc., *Control System Toolbox Reference*, Natick, MA, 2025.